

Questions with Answer Keys

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Q1 - 2024 (01 Feb Shift 1)

A monochromatic light of wavelength $6000\text{\AA}$ is incident on the single slit of width 0.01 mm . If the diffraction pattern is formed at the focus of the convex lens of focal length 20 cm , the linear width of the central maximum is :

- (1) 60 mm
- (2) 24 mm
- (3) 120 mm
- (4) 12 mm

Q2 - 2024 (01 Feb Shift 2)

A microwave of wavelength 2.0 cm falls normally on a slit of width 4.0 cm . The angular spread of the central maxima of the diffraction pattern obtained on a screen 1.5 m away from the slit, will be:

- (1) 30°
- (2) 15°
- (3) 60°
- (4) 45°

Q3 - 2024 (01 Feb Shift 2)

In Young's double slit experiment, monochromatic light of wavelength $5000\text{\AA}$ is used. The slits are 1.0 mm apart and screen is placed at 1.0 m away from slits. The distance from the centre of the screen where intensity becomes half of the maximum intensity for the first time is $\underline{\hspace{1cm}} \times 10^{-6}\text{ m}$.

Q4 - 2024 (27 Jan Shift 2)

When a polaroid sheet is rotated between two crossed polaroids then the transmitted light intensity will be maximum for a rotation of :

- (1) 60°
- (2) 30°

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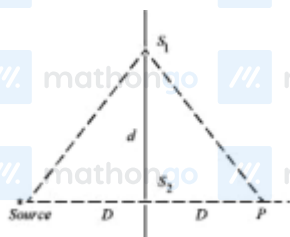
(3) 90° (4) 45°

Q5 - 2024 (27 Jan Shift 2)

A parallel beam of monochromatic light of wavelength $5000 \text{ \AA}$ is incident normally on a single narrow slit of width 0.001 mm . The light is focused by convex lens on screen, placed on its focal plane. The first minima will be formed for the angle of diffraction of _____ (degree).

Q6 - 2024 (29 Jan Shift 1)

In a double slit experiment shown in figure, when light of wavelength 400 nm is used, dark fringe is observed at P . If $D = 0.2 \text{ m}$, the minimum distance between the slits S_1 and S_2 is _____ mm.



Q7 - 2024 (29 Jan Shift 2)

In Young's double slit experiment, light from two identical sources are superimposing on a screen. The path difference between the two lights reaching at a point on the screen is $\frac{7\lambda}{4}$. The ratio of intensity of fringe at this point with respect to the maximum intensity of the fringe is :

(1) $1/2$ (2) $3/4$ (3) $1/3$ (4) $1/4$

Q8 - 2024 (29 Jan Shift 2)

In a single slit diffraction pattern, a light of wavelength $6000 \text{ \AA}$ is used. The distance between the first and third minima in the diffraction pattern is found to be 3 mm when the screen is placed 50 cm away from slits. The

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width of the slit is _____ $\times 10^{-4}$ m.

Q9 - 2024 (30 Jan Shift 1)

The diffraction pattern of a light of wavelength 400 nm diffracting from a slit of width 0.2 mm is focused on the focal plane of a convex lens of focal length 100 cm. The width of the 1st secondary maxima will be :

- (1) 2 mm
- (2) 2 cm
- (3) 0.02 mm
- (4) 0.2 mm

Q10 - 2024 (30 Jan Shift 2)

A beam of unpolarised light of intensity I_0 is passed through a polaroid A and then through another polaroid B which is oriented so that its principal plane makes an angle of 45° relative to that of A. The intensity of emergent light is :

- (1) $I_0/4$
- (2) I_0
- (3) $I_0/2$
- (4) $I_0/8$

Q11 - 2024 (31 Jan Shift 1)

Two waves of intensity ratio 1 : 9 cross each other at a point. The resultant intensities at the point, when (a)

Waves are incoherent is I_1 (b) Waves are coherent is I_2 and differ in phase by 60° . If $\frac{I_1}{I_2} = \frac{10}{x}$ then x = _____

Q12 - 2024 (31 Jan Shift 2)

When unpolarized light is incident at an angle of 60° on a transparent medium from air. The reflected ray is completely polarized. The angle of refraction in the medium is

- (1) 30°

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(2) 60°						
(3) 90°						
(4) 45°						

Questions with Answer Keys

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Answer Key

Answer Key

Q1 (2) mathongo **Q2 (3)** mathongo **Q3 (125)** mathongo **Q4 (4)** mathongo

Q5 (30)  **Q6 (0.2)**  **Q7 (1)**  **Q8 (2)** 

Q9 (1) mathongo // mathongo **Q10** (1) // mathongo **Q11** (13) mathongo // mathongo **Q12** (1) // mathongo

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Solutions

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Q1

Linear width

$$W = \frac{2\lambda d}{a} = \frac{2 \times 6 \times 10^{-7} \times 0.2}{1 \times 10^{-5}} = 2.4 \times 10^{-2} = 24 \text{ mm}$$

Q2

For first minima $a \sin \theta = \lambda$

$$\sin \theta = \frac{\lambda}{a} = \frac{1}{2}$$

$$\theta = 30^\circ$$

$$\text{Angular spread} = 60^\circ$$

Q3

Let intensity of light on screen due to each slit is I_0 So intensity at centre of screen is $4I_0$ Intensity at distance y from centre-

$$I = I_0 + I_0 + 2\sqrt{I_0 I_0} \cos \phi$$

$$I_{\max} = 4I_0$$

$$\frac{I_{\max}}{2} = 2I_0 = 2I_0 + 2I_0 \cos \phi$$

$$\cos \phi = 0$$

$$\phi = \frac{\pi}{2}$$

$$K\Delta x = \frac{\pi}{2}$$

$$\frac{2\pi}{\lambda} d \sin \theta = \frac{\pi}{2}$$

$$\frac{2}{\lambda} d \times \frac{y}{D} = \frac{1}{2}$$

$$y = \frac{\lambda D}{4d} = \frac{5 \times 10^{-7} \times 1}{4 \times 10^{-3}}$$

$$= 125 \times 10^{-6}$$

$$= 125$$

Q4

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Solutions

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Let I_0 be intensity of unpolarised light incident on first polaroid.

$$I_1 = \text{Intensity of light transmitted from 1}^{\text{st}} \text{ polaroid}$$

$$= \frac{I_0}{2}$$

θ be the angle between 1st and 2nd polaroid

ϕ be the angle between 2nd and 3rd polaroid

$\theta + \phi = 90^\circ$ (as 1st and 3rd polaroid are crossed)

$$\phi = 90^\circ - \theta$$

I_2 = Intensity from 2nd polaroid

$$I_2 = I_1 \cos^2 \theta = \frac{I_0}{2} \cos^2 \theta$$

I_3 = Intensity from 3rd polaroid

$$I_3 = I_2 \cos^2 \phi$$

$$I_3 = I_1 \cos^2 \theta \cos^2 \phi$$

$$I_3 = \frac{I_0}{2} \cos^2 \theta \cos^2 \phi$$

$$\phi = 90 - \theta$$

$$I_3 = \frac{I_0}{2} \cos^2 \theta \sin^2 \theta$$

$$I_3 = \frac{I_0}{2} \left[\frac{2 \sin \theta \cos \theta}{2} \right]^2$$

$$I_3 = \frac{I_0}{8} \sin^2 2\theta$$

I_3 will be maximum when $\sin 2\theta = 1$

$$2\theta = 90^\circ$$

$$\theta = 45^\circ$$

Q5

For first minima

$$a \sin \theta = \lambda$$

$$\Rightarrow \sin \theta = \frac{\lambda}{a} = \frac{5000 \times 10^{-10}}{1 \times 10^{-6}} = \frac{1}{2}$$

$$\Rightarrow \theta = 30^\circ$$

Q6

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Solutions

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Path difference for minima at P

$$2\sqrt{D^2 + d^2} - 2D = \frac{\lambda}{2}$$

$$\therefore \sqrt{D^2 + d^2} - D = \frac{\lambda}{4}$$

$$\therefore \sqrt{D^2 + d^2} = \frac{\lambda}{4} + D$$

$$\Rightarrow D^2 + d^2 = D^2 + \frac{\lambda^2}{16} + \frac{D\lambda}{2}$$

$$\Rightarrow d^2 = \frac{D\lambda}{2} + \frac{\lambda^2}{16}$$

$$\Rightarrow d^2 = \frac{0.2 \times 400 \times 10^{-9}}{2} + \frac{4 \times 10^{-14}}{4}$$

$$\Rightarrow d^2 \approx 400 \times 10^{-10}$$

$$\therefore d = 20 \times 10^{-5}$$

$$\Rightarrow d = 0.20 \text{ mm}$$

Q7

$$\Delta x = \frac{7\lambda}{4}$$

$$\phi = \frac{2\pi}{\lambda} \Delta x = \frac{2\pi}{\lambda} \times \frac{7\lambda}{4} = \frac{7\pi}{2}$$

$$I = I_{\max} \cos^2\left(\frac{\phi}{2}\right)$$

$$\frac{I}{I_{\max}} = \cos^2\left(\frac{\phi}{2}\right) = \cos^2\left(\frac{7\pi}{2 \times 2}\right) = \cos^2\left(\frac{7\pi}{4}\right)$$

$$= \cos^2\left(2\pi - \frac{\pi}{4}\right)$$

$$= \cos^2 \frac{\pi}{4}$$

$$= \frac{1}{2}$$

Q8

For n^{th} minima

$$b \sin \theta = n\lambda$$

(λ is small so $\sin \theta$ is small, hence $\sin \theta \simeq \tan \theta$)

$$b \tan \theta = n\lambda$$

$$b \frac{y}{D} = n\lambda$$

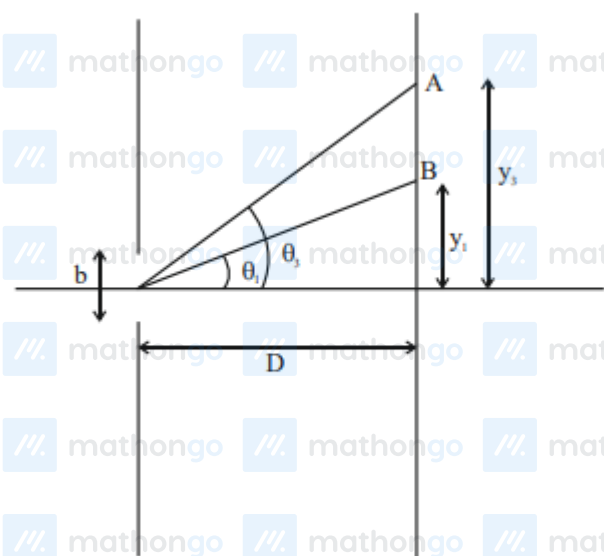
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Solutions

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$$\Rightarrow y_n = \frac{n\lambda D}{b} \text{ (Position of } n^{\text{th}} \text{ minima)}$$



B $\rightarrow$ 1st minima, A $\rightarrow$ 3rd minima

$$y_3 = \frac{3\lambda D}{b}, y_1 = \frac{\lambda D}{b}$$

$$\Delta y = y_3 - y_1 = \frac{2\lambda D}{b}$$

$$3 \times 10^{-3} = \frac{2 \times 6000 \times 10^{-10} \times 0.5}{b}$$

$$b = \frac{2 \times 6000 \times 10^{-10} \times 0.5}{3 \times 10^{-3}}$$

$$b = 2 \times 10^{-4} \text{ m}$$

$$x = 2$$

Q9

$$\text{Width of 1}^{\text{st}} \text{ secondary maxima} = \frac{\lambda}{a} \cdot D$$

Here

$$a = 0.2 \times 10^{-3} \text{ m}$$

$$\lambda = 400 \times 10^{-9} \text{ m}$$

$$D = 100 \times 10^{-2}$$

Width of 1st secondary maxima

$$= \frac{400 \times 10^{-9}}{0.2 \times 10^{-3}} \times 100 \times 10^{-2}$$

$$= 2 \text{ mm}$$

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Solutions

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Q10

Intensity of emergent light

$$= \frac{I_0}{2} \cos^2 45^\circ = \frac{I_0}{4}$$

Q11

For incoherent wave $I_1 = I_A + I_B \Rightarrow I_1 = I_0 + 9I_0$

$$I_1 = 10I_0$$

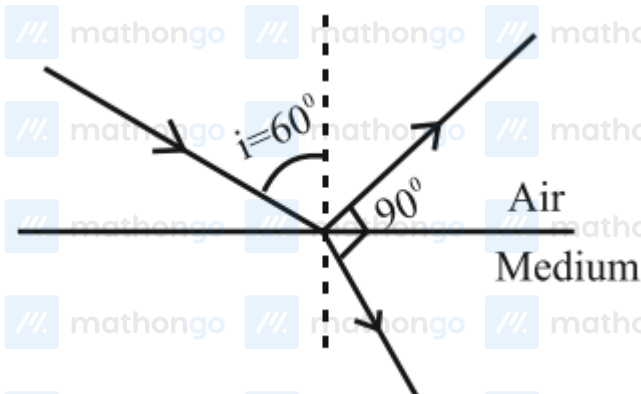
For coherent wave $I_2 = I_A + I_B + 2\sqrt{I_A I_B} \cos 60^\circ$

$$I_2 = I_0 + 9I_0 + 2\sqrt{9I_0^2} \cdot \frac{1}{2} = 13I_0$$

$$\frac{I_1}{I_2} = \frac{10}{13}$$

Q12

By Brewster's law



At complete reflection refracted ray and reflected ray are perpendicular.